Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/18/2025
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	01/01/2022	**End Date:**	09/30/2027
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2027 per D. Stocker/GRC (Ed., 9/19/25). NOTE: End date changed to 09/30/2025 per NSSC information (Ed., 12/29/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Carey, Van Ph.D. (University of California, Berkeley)		
Grant/Contract No.:	80NSSC22K0329		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A program to study the flammability characteristics of wiring materials in spacecraft environments (that will complement the existing Japanese Space Agency (JAXA) project entitled, “Fundamental Research on International Standard of Fire Safety in Space”) is being carried out at U.C. Berkeley under NASA sponsorship, with the objective of providing information about electrical wires flammability in “Space Exploration Atmospheres” (SEA). The wires under study are similar to those of the tests being conducted at Hokkaido University by the Japanese counterpart researchers, and the tests are planned to be conducted in the International Space Station/Japanese Experiment Module (ISS/KIBO) unit. The long-term goal of the project is to understand the changes in electrical wire flammability, specifically the flame spread Limiting Oxygen Index - Maximum Oxygen Concentration (LOI-MOC), in the environment expected in space-based facilities – in particular, those at space exploration atmospheres (SEA). These environments include reduced gravity, low velocity flows, reduced pressures, and elevated oxygen concentrations. The near-term objective of the research is to conduct tests of wire burning in a ground-based Flammability Limits at Reduced-g Experiment (FLARE) UCB apparatus at varied flow velocities, reduced ambient pressure, and elevated oxygen concentration. Another objective of the project is to model the experimental data using Neural Networks to study the effectiveness of that approach in predicting wire burning behavior in spacecraft environments. The experiments and modeling support the ISS/KIBO experiments to be conducted under the FLARE/JAXA project. The project experimental methodology consists of obtaining flame spread rates at varied flow velocity and ambient pressure and oxygen concentration in wires with different cores and insulation materials (primarily those being studied at Hokkaido University and planned for the tests in the ISS/KIBO). The wires under study are primarily “laboratory” type wires consisting of a conducting core, primary copper, and a polymeric insulation (primarily PE). The UCB experiments are being conducted in normal gravity, with a limited number of tests conducted in an aircraft following a parabolic path. The testing methodology is supported with theoretical modeling using Neural Network and Genetic Algorithms to predict flame spread over wires at different geometrical and ambient conditions. This research primarily supports the FLARE/JAXA program but also could lead to better determination and ranking of the fire hazard characteristics of potential wire materials to be used in spacecraft for long-term exploration missions. NOTE this is continuation of "Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space," grant #80NSSC22K0329 with the same principal investigator, Prof. Carlos Fernandez-Pello. Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in spacecraft allows us to accurately elucidate the effect of the environment parameters on the ignition and flame spread over combustible material, and through them their potential fire hazard. Particularly important is the determination of the Limiting Oxygen Concentration (LOC) on flame extinction under spacecraft environments. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Another important aspect of the research is the effect of melting and dripping of plastic insulation in normal gravity in comparison with microgravity. The results are relevant because dripping will not occur in microgravity and consequently could impact their burning and methodology to screen. The investigation and results have also benefits for terrestrial fire safety by providing further information about the flammability of materials under a variety of environments.
	A summary of relevant research progress during this reporting period (January 1, 2024-February 28, 2025) is presented here. The research includes a theoretical and an experimental task. The theoretical task consisted primarily in developing Machine Learning Algorithms using the available experimental data of the flame spread over insulated wires. The experimental task consisted of an initial microgravity test of the spread of flames over an LDPE/Cu wire, conducted in the Kibo unit of the International Space Station (ISS). Although more microgravity experiments were planned during the current reporting period, a malfunction in the mechanism that changes the samples to be tested only permitted one initial test. The astronauts were not able to solve the problem, and the apparatus had to be returned to Earth for repair. The new apparatus has already been sent to the ISS and new tests are being planned as soon as possible. More detailed information about the research conducted can be found in the cited publications. Theoretical task: Advancing on the prediction of flame spread over electrical wires predictions by using Machine Learning with data categorization and augmentation strategies Flame spread over electrical wires is influenced by a variety of factors, including core and insulation materials, geometric dimensions, and environmental conditions. Traditional modeling approaches struggle to capture the complexity and non-linearity arising from these intertwined parameters. In this study, we combine machine learning (ML) algorithms with data categorization and augmentation strategies to improve predictive accuracy and interpretability of flame spread rate (FSR) models. We begin with a comprehensive dataset compiled from literature, encompassing a broad range of input parameters. A baseline is established using raw numerical data, which yields low to moderate predictive capability. Introducing categorical variables such as type of insulation and flame spread direction by using the One-Hot encoding technique improves the predictive capabilities of the models. This approach transformed classification features into binary indicators, allowing algorithms to leverage discrete categories more effectively unveiling hidden patterns, enhancing the model predictive capabilities. Non-linear models (e.g., Decision Trees, Gradient Boosting, Multi-Layer Perceptron) achieve R^2 values above 0.9. Data augmentation methods, including bootstrap resampling and Generative Adversarial Networks (GANs), further refine performance by filling data gaps and providing more robust training conditions. In practice, systematic measurements of all parameters necessary to predict flame spread over wires insulation are not feasible, and the data reported in the literature are often non-uniform. This lack of comprehensive datasets hampers direct validation of theoretical models and limits their predictive capabilities. Thus, while existing correlations capture some essential physics of the problem, they fall short of incorporating the full range of parameters observed in experiments. To address these challenges, this work introduces alternative methodologies that integrate ML and data augmentation

techniques to enhance predictive modeling capabilities for flame spread rates over electrical wires. Ultimately, this study explores methods that guide future research towards more effectively incorporating physical principles into the machine learning prediction process during the training phase. The contribution of this study lies not only in applying ML to a traditionally physics-driven domain, but also in introducing data augmentation methods—such as bootstrapping, noise injection, and the use of Generative Adversarial Networks (GANs). Although GANs are typically utilized for image-based data, here they are adapted and implemented to expand training datasets for the FSR problem artificially. By using data augmentation, the Machine Learning algorithms can better capture the underlying distributions and physical behaviors reflected in real experimental observations. This, in turn, can improve prediction accuracy, enhance model generalization, and potentially yield deeper insights into the relative importance of various input parameters.

Task Progress:

Experimental task: Microgravity experiments in the ISS

The primary objective of the microgravity experiments is to provide information about the flame spread and extinction of simulated electrical wires in atmospheres expected in futures spacecraft. Experiments with laboratory wires consisting of a Cu core and Low-Density Polyethylene (LDPE) insulation samples are conducted in a sealed combustion chamber in the Kibo unit of the ISS, and in a similar chamber in normal gravity. The characteristics of the spread of flames over the LDPE wires at different opposed flow velocities are observed at different ambient pressures and oxygen concentrations. In the experiments, the differences in melting and dripping of the insulation in normal and microgravity are observed to understand the effect of gravity on the wire burning. The path to extinction of the opposed spread of flames is also followed and their limiting oxygen volume fraction at extinction (LOC) recorded. Normal gravity experiments were conducted and reported in previous reporting periods. In this report, only the microgravity experiments are reported.

The microgravity tests were performed in the SCEM/FLARE apparatus developed at the Japanese Space Agency (JAXA) and that is located in the Kibo module of the International Space Station. [Ed. Note: FLARE is the Flammability Limits at Reduced-g Experiment.] The flame spread behavior over a test specimen can be captured by three visible cameras and one infrared camera. The SCEM/FLARE has a circulating wind tunnel structure that allows adjustment of total pressure, oxygen concentration, and oxidizer flow velocity in the test section. A test specimen is ignited by a hot wire igniter, and the opposed-flame spread along the test specimen is observed. All the tests are conducted with “laboratory” wires consisting of 125mm-long copper cores and 100mm-long LDPE insulation sheaths; the copper cores had diameters of 0.64 mm, and the insulation sheaths had inner diameters of 0.7 mm and outer diameters of 4 mm. The relatively large size of the “laboratory” wires and the insulation material, while not intended to necessarily reproduce actual electrical wires, were selected to facilitate the interpretation of the experimental results. For the study, the wire samples placed in the middle of a flow duct were all contained within an air-tight pressure vessel with access to fresh air and the ability to remove exhaust to keep the experimental environment consistent throughout the length of each test.

On August 6, 2024, a combustion test was conducted on the Japanese Experiment Module Kibo of the ISS, of the opposed flow spread of flames over a Cu/LDPE wire. Although initially, twelve combustion tests were planned in 2024, due to a malfunction of the sample exchange system, only one combustion test could be conducted. The opposed flow flame spread test was conducted with the chamber at an O2 concentration of 25%, and an opposed flow velocity = 10 cm/s. After the ignition of the LDPE insulation is established, a flame starts spreading along the solid fuel surface against the opposed forced gas flow. Various interesting phenomena were observed during the flame spreading process, such as the deformation of the molten LDPE and the transition from flame flash to a flamelet spreading. The flame spread rate was smaller than in normal gravity and the flame reached the end of the example without extinction. Although the extinction limit was not determined, it was found that the flame spread at lower oxygen concentration than in normal gravity. The differences in melting, and no dripping, in microgravity and normal gravity were also notable, and influenced the flame significantly the flame spread process under both conditions.

The motion history of the flame was analyzed from sequentially numbered visual images taken by the VC2 camera. The time history of the flame spread rate was calculated from the motion history of the upstream end of the flame by using a SG filter. The time at which a flash was observed was confirmed from the motion history of the downstream end of the flame. After the flash was observed, the flame changes to a flamelet, and its spread rate is almost constant at 1.4 mm/s.

Bibliography Type: Description: (Last Updated: 02/15/2026)

Abstracts for Journals and Proceedings

Scudiere C, Gagnon L, Carey V, Fernandez-Pello C. "Modeling of flame spread along wire utilizing an artificial neural network and a genetic algorithm." 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023
Abstracts. 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023 , Mar-2023

Abstracts for Journals and Proceedings

Rivera J, San Martin D, Fernandez-Pello C, Gollner M, Olson. S. J. Rivera, D. San Martin, C. Fernandez-Pello, M. Gollner, S. Olson. "Advancing on the flame spread over electrical wires predictions by using machine learning with data categorization and augmentation strategies." 11th International Seminar on Fire and Explosion Hazards (ISFEH), Rome, Italy, June 15-21, 2025.
Abstracts. 11th International Seminar on Fire and Explosion Hazards (ISFEH), Rome, Italy, June 15-21, 2025. , Jun-2025